

Math Notebook

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1 Analysis

1.1 Conventions and notation

1.1.1 Abbreviations for references

- TL: Kalkulus - Tom Lindstrøm - 4. udgave
- An1: Analyse 1 - Matthias Christandl, Søren Eilers og Henrik Schlichtkrull - Syvende udgave

1.1.2 Standard number sets

Notation of standard sets of numbers in the analysis section of this notebook.

- $\mathbb{N} = \{1, 2, 3, \dots\}$
- $\mathbb{N}_0 = \{0, 1, 2, \dots\}$
- $\mathbb{Z}^+ = \mathbb{N}$
- $\mathbb{R}^+ = \{x \in \mathbb{R} \mid x > 0\}$

1.1.3 Placeholder sets

Unless otherwise stated, uppercase Latin letters $A, B, C, \dots$ denote subsets of $\mathbb{R}$. I.e. whenever something like $f : A \rightarrow B$ is written without further context of the domain and codomain A and B , assume that $A, B \subseteq \mathbb{R}$.

This assumption also holds for functions defined like $f : D_f \rightarrow V_f$.

1.1.4 Mathematical statements $P(x)$ or $P(n)$

Not to be confused with the power set $\mathcal{P}(S)$, the notation $P(x)$ or $P(n)$ refers to a mapping $P : D_P \rightarrow \{\perp, \top\}$ that assigns a truth value based on whether the statement holds for a given input value.¹ In this notebook, unless otherwise stated, assume that $D_P \subseteq \mathbb{R}$ when $P(x)$ is written, and assume that $D_P \subseteq \mathbb{N}$ when $P(n)$ is written.

1.1.5 Explicit dependence e.g. $N = N(\varepsilon)$, $\delta = \delta(\varepsilon)$

When you see something like " $\varepsilon \in \mathbb{R}^+$, $\delta = \delta(\varepsilon)$ " in this notebook, you may read it as "Let ε be a positive real number, and let δ be a number where its value depends in some way on the value of ε ". This notation is simply for ease of understanding the structure of proofs and theorems, and may show itself in epsilon-N proofs (for sequences) like

$$\forall \varepsilon \in \mathbb{R}^+ \exists N = N(\varepsilon) \in \mathbb{N} \text{ s.t. } n > N \implies |a_n - a| < \varepsilon \quad (1.1.5.1)$$

¹The $\{\perp, \top\}$ is equivalent to $\{\text{False}, \text{True}\}$, $\{0, 1\}$, etc.

or epsilon-delta proofs like

$$\forall \varepsilon \in \mathbb{R}^+ \exists \delta = \delta(\varepsilon) \in \mathbb{R}^+ \text{ s.t. } 0 < |x - a| < \delta \implies |f(x) - b| < \varepsilon \quad (1.1.5.2)$$

In a similar manner, if $y \in A$ is defined as an expression containing $x \in B$, one may also write $y = y(x)$ since y depends on the value of x .

1.2 Given definitions and prerequisites

The following concepts are assumed known:

- Basic set theory notation
- Real number arithmetic
- Functions and mappings
- Mathematical induction
- Proof techniques

1.2.1 Sequences

A sequence is a function whose domain is an ordered subset of the integers (usually $\mathbb{N}$). If we let $n \in \mathbb{N}$ and $a_n \in \mathbb{R}$, then $(a_n)_{n \in \mathbb{N}} = (a_1, a_2, a_3, \dots)$ is said to be a real sequence. Each image² a_n is called a term of the sequence.

One can view a sequence as simply being a map $a : D_a \rightarrow V_a$. The codomain may consist of any set that contains the sequence's possible outputs, for example, $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, or even vector spaces like $\mathbb{R}^2$ or $\mathbb{C}^2$. A sequence $(a_n)_{n \in A}$ may also be finite over a set $A = \{n_1, n_2, \dots, n_k\}$ with $k \in \mathbb{N}$. Unless otherwise stated though, assume that any given sequence is a map $a : \mathbb{N} \rightarrow \mathbb{R}$.

A finite sequence of length two may be denoted as (a, b) . To avoid confusing it with the notation of an open interval, assume that (a, b) refers to an open interval if it is not stated to be a finite sequence.

1.2.2 Series

A series is an expression representing the sum of the terms in a finite or infinite sequence. For $(a_n)_{n \in \mathbb{N}}$ we define its corresponding series as the expression

$$\sum_{n=1}^{\infty} a_n \quad (1.2.2.1)$$

²An image is the value of a function evaluated at a specific input

A partial sum, S_n , of the series $\sum_{k=1}^{\infty} a_k$ is the sum of the terms within a finite subsequence $(a_k)_{k \in A}$ with $A = \{k \in \mathbb{N} \mid 1 \leq k \leq n\}$.³ The n -th partial sum S_n may then be defined as

$$S_n = \sum_{k=1}^n a_k \quad (1.2.2.2)$$

1.2.2.1 Limit of partial sum

The limit of the partial sum S_n converges to $S \in \mathbb{R}$ as n approaches infinity if and only if

$$\forall \varepsilon \in \mathbb{R}^+ \exists N = N(\varepsilon) \in \mathbb{N} \text{ s.t. } n > N \implies |S_n - S| < \varepsilon \quad (1.2.2.3)$$

i.e.

$$\lim_{n \rightarrow \infty} S_n = S \quad (1.2.2.4)$$

which means that the series converges to the value S :

$$\sum_{n=1}^{\infty} a_n = S \quad (1.2.2.5)$$

1.2.3 Limits over real functions

Symbolic definition of limit for $f : \mathbb{R} \rightarrow \mathbb{R}$

$$\forall \varepsilon \in \mathbb{R}^+ \exists \delta = \delta(\varepsilon) \in \mathbb{R}^+ \text{ s.t. } \forall x \in D_f : 0 < |x - a| < \delta \implies |f(x) - b| < \varepsilon \quad (1.2.3.1)$$

This defines the standard limit notation of

$$\lim_{x \rightarrow a} f(x) = b \quad (1.2.3.2)$$

Definition is taken from TL definition 5.4.1.

1.2.4 Continuity

A function $f : D_f \rightarrow V_f$ is said to be continuous at a point $a \in D_f$ if the following holds:

$$\forall \varepsilon \in \mathbb{R}^+ \exists \delta = \delta(\varepsilon) \in \mathbb{R}^+ \text{ s.t. } \forall x \in D_f : |x - a| < \delta \implies |f(x) - f(a)| < \varepsilon \quad (1.2.4.1)$$

Definition is taken from TL definition 5.1.1.

1.2.5 Derivatives

Given a function $f : D_f \rightarrow V_f$, the derivative of f at a point $a \in D_f$ is defined as

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \quad (1.2.5.1)$$

Or equivalently

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad (1.2.5.2)$$

³Pretentious way of writing $A = \{1, 2, 3, \dots, n\}$

1.3 Derivative properties

1.3.1 Sum rule

Let $f : A \rightarrow B$ be a function consisting of the sum of n functions, where each function is a term in the sequence $(f_k(x))_{k \leq n} = (f_1(x), f_2(x), \dots, f_n(x))$:

$$f(x) = \sum_{k=1}^n f_k(x) \quad (1.3.1.1)$$

Sum rule:

$$\left(\sum_{k=1}^n f_k(x) \right)' = \sum_{k=1}^n f'_k(x) \quad (1.3.1.2)$$

Proof by induction:

$$P(n) := \left(\sum_{k=1}^n f_k(x) \right)' = \sum_{k=1}^n f'_k(x) \quad (1.3.1.3)$$

Base case⁴ of $P(n)$ for $n = 2$:

$$\begin{aligned} f(x) &= f_1(x) + f_2(x) \\ \implies f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(f_1(x+h) + f_2(x+h)) - (f_1(x) + f_2(x))}{h} \\ &= \left(\lim_{h \rightarrow 0} \frac{f_1(x+h) - f_1(x)}{h} \right) + \left(\lim_{h \rightarrow 0} \frac{f_2(x+h) - f_2(x)}{h} \right) \\ &= f'_1(x) + f'_2(x) \end{aligned}$$

By definition 1.2.5

⁴Base case is actually 1 for this proof to be valid, but that's trivial and boring so I wanna do 2 instead.

Induction step:

$$\begin{aligned}
\left((P(2) \wedge P(n)) \right) &\implies \left(f(x) = \sum_{k=1}^{n+1} f_k(x) \right. \\
&= f_{n+1}(x) + \sum_{k=1}^n f_k(x) \\
&\implies f'(x) = \left(f_{n+1}(x) + \sum_{k=1}^n f_k(x) \right)' \\
&= f'_{n+1}(x) + \left(\sum_{k=1}^n f_k(x) \right)' && \text{By } P(2) \\
&= f'_{n+1}(x) + \sum_{k=1}^n f'_k(x) && \text{By } P(n) \\
&= \sum_{k=1}^{n+1} f'_k(x) \\
&\implies P(n+1)
\end{aligned}$$

■

1.3.2 Power rule

Let $f : A \rightarrow B$ be a function of the form $f(x) = x^n$. The power rule thus states that the derivative of f is the following

$$f'(x) = (x^n)' = nx^{n-1} \quad (1.3.2.1)$$

Proof

$$\begin{aligned}
(x^n)' &= \lim_{h \rightarrow 0} \frac{(x+h)^n - x^n}{h} \\
&= \lim_{h \rightarrow 0} \frac{-x^n + \sum_{k=0}^n \binom{n}{k} x^{n-k} h^k}{h} \\
&= \lim_{h \rightarrow 0} \frac{\sum_{k=1}^n \binom{n}{k} x^{n-k} h^k}{h} \\
&= \lim_{h \rightarrow 0} \sum_{k=1}^n \binom{n}{k} x^{n-k} h^{k-1} \\
&= \lim_{h \rightarrow 0} \binom{n}{1} x^{n-1} h^0 + \sum_{k=2}^n \binom{n}{k} x^{n-k} h^{k-1} \\
&= nx^{n-1} + \lim_{h \rightarrow 0} \sum_{k=2}^n \binom{n}{k} x^{n-k} h^{k-1} \\
(h \rightarrow 0 \wedge k-1 \geq 1) &\implies (x^n)' = nx^{n-1}
\end{aligned}$$

■

1.3.3 Constant coefficient rule

For any function $f : A \rightarrow B$ and a constant c , we have that

$$(cf(x))' = cf'(x) \quad (1.3.3.1)$$

Proof

$$\begin{aligned} (cf(x))' &= \lim_{h \rightarrow 0} \frac{cf(x+h) - cf(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{c(f(x+h) - f(x))}{h} \\ &= c \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= cf'(x) \end{aligned} \quad \blacksquare$$

1.3.4 Product rule

For any two functions f and g , we have that

$$(f(x)g(x))' = f(x)g'(x) + f'(x)g(x) \quad (1.3.4.1)$$

1.4 Derivatives of standard functions

1.4.1 Polynomials

1.4.1.1 Degree-1 polynomials

Let a function $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = x$. For the derivative for a fixed point $a \in \mathbb{R}$, we have from equation (1.2.5.1) that

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{x \rightarrow a} \frac{x - a}{x - a} = \lim_{x \rightarrow a} \frac{1}{1} = 1 \quad (1.4.1.1)$$

Redefine f to the general form $f(x) = mx + b$. We then have

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{x \rightarrow a} \frac{mx + b - ma - b}{x - a} = \lim_{x \rightarrow a} \frac{m(x - a)}{x - a} = \lim_{x \rightarrow a} \frac{m}{1} = m \quad (1.4.1.2)$$

1.5 L'Hôpital's rule

1.5.1 Motivation

1.5.2 Statement

1.5.3 Required lemmas

1.5.4 Proof

1.5.5 Examples of L'Hôpital's rule